In[•]:=
```
(*phase_kinetics.m:SO(4,4) 3D Bipartite Möbius Braid*)
Manipulate[Module[{bulkSphere, matterTrace, antimatterTrace, k},
  (*Using a half-integer frequency (freq/2) forces the Möbius topology.A full 4π
    rotation is required to close the loop,splitting naturally into two 2π out-
   of-phase halves (Matter and Antimatter).*)k = freq / 2;
  (*Parametric equations for the wave wrapped around the bulk sphere*)
  x[t_] := (bulkRadius + waveAmplitude * Cos[braidTwist * t]) * Cos[t] * Cos[k * t];
  y[t_] := (bulkRadius + waveAmplitude * Cos[braidTwist * t]) * Sin[t] * Cos[k * t];
  z[t_] := (bulkRadius + waveAmplitude * Cos[braidTwist * t]) * Sin[k * t];
  (*The central 6D/8D Bulk (Translucent)*)
  bulkSphere = Graphics3D[{Opacity[0.3], LightGray, Sphere[{0, 0, 0}, bulkRadius]}];
  (*The First 2π Cycle:Matter (+mi/Real Dominant)*)
  matterTrace = ParametricPlot3D[{x[t], y[t], z[t]}, {t, 0, 2 Pi},
    PlotStyle → Directive[Thickness[0.015], Blue, Specularity[White, 50]],
    PlotLegends → {"Matter Wave (+mi Phase)"}];
  (*The Second 2π Cycle:Antimatter (-mi/Imaginary Dominant)*)
  antimatterTrace = ParametricPlot3D[{x[t], y[t], z[t]}, {t, 2 Pi, 4 Pi},
    PlotStyle → Directive[Thickness[0.015], Red, Specularity[White, 50]],
    PlotLegends → {"Antimatter Wave (-mi Phase)"}];
  (*Combine and render the entangled geometry*)
  Show[bulkSphere, matterTrace, antimatterTrace, PlotRange → All,
   Boxed → False, Axes → False, Background → Black, ImageSize → Large,
   PlotLabel → Style["SO(4,4) Spin-Paired Bipartite Braid", White, 14, Bold]]],
 (*Topology and Harmonic Sliders*)
 {{freq, 7, "Harmonic Nodes (Must be Odd for Möbius Twist)"}, 1, 15, 2},
 {{bulkRadius, 3.0, "Radius of the Inner Bulk"}, 1.0, 5.0},
 {{waveAmplitude, 0.5, "Amplitude of the Metric Wave"}, 0.1, 2.0},
 {{braidTwist, 2.0, "Secondary Double-Dutch Braid Twist"}, 0, 5}]
```

Out[ ]=

Harmonic Nodes (Must be Odd for Möbius Twist)

Radius of the Inner Bulk

Amplitude of the Metric Wave

Secondary Double –Dutch Braid Twist

